Cryptocurrencies and ICOs

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Cryptocurrencies: Market Overview

The Crypto Market: Scale and Volatility

Total Market Cap: peaked at $3 trillion (Nov 2021); crashed to $800B (Dec 2022 – FTX collapse); recovered to $2.5T+ by early 2024

Bitcoin Dominance: consistently 45–55% of total crypto market cap; rises in bear markets as capital flees altcoins

Asset Class Correlation: crypto correlates with risk assets (Nasdaq r≈0.6 in 2022); decoupling observed in 2024 bull run driven by ETF demand

Retail vs. Institutional: 2017 was retail-driven; 2024 institutions hold 25%+ of Bitcoin supply via ETFs and corporate treasuries

Jan. 1 2025

Rank	Name	Symbol	Market Cap	Price
1	Bitcoin	BTC	$1,850,264,551,569.93	$93,429.20
2	Ethereum	ETH	$401,479,034,540.14	$3,332.53
3	Tether USDt	USDT	$137,178,559,146.95	$0.9981
4	XRP	XRP	$119,420,603,990.73	$2.0801
5	BNB	BNB	$100,947,038,231.85	$700.99
6	Solana	SOL	$91,382,941,466.73	$189.26
7	Dogecoin	DOGE	$46,542,481,541.87	$0.3157
8	USDC	USDC	$43,944,859,141.92	$0.9999
9	Cardano	ADA	$29,649,723,778.26	$0.8438
10	TRON	TRX	$21,911,966,118.92	$0.2542

Jan. 1 2024

Rank	Name	Symbol	Market Cap	Price
1	Bitcoin	BTC	$827,811,209,384.41	$42,265.19
2	Ethereum	ETH	$274,194,287,336.35	$2,281.47
3	Tether USDt	USDT	$91,674,848,734.73	$0.9997
4	BNB	BNB	$47,394,056,692.15	$312.44
5	Solana	SOL	$43,576,016,359.83	$101.51
6	XRP	XRP	$33,283,785,350.85	$0.6149
7	USDC	USDC	$24,519,751,149.21	$1.0001
8	Cardano	ADA	$21,014,927,330.99	$0.5942
9	Avalanche	AVAX	$14,086,866,595.76	$38.54
10	Dogecoin	DOGE	$12,746,626,943.45	$0.08947

TOP 15 STABLECOINS BY MARKET CAPITALIZATION

	NAME	MARKET CAP	1M MKT CAP CHANGE
1	TETHER [USDT]	$ 65.3B	-5.3%
2	USD COIN [USDC]	$ 44.3B	+1.4%
3	BINANCE USD [BUSD]	$ 22.4B	+4.7%
4	DAI [DAI]	$ 5.19B	-10.0%
5	FRAX [FRAX]	$ 1.18B	-3.3%
6	PAX DOLLAR [USDP]	$ 910M	+1.4%
7	TRUE USD [TUSD]	$ 807M	-1.8%
8	USDD [USDD]	$ 718M	-0.8%
9	GEMINI DOLLAR [GUSD]	$ 602M	+25.4%
10	PAX GOLD [PAXG]	$ 477M	-4.2%
11	TETHER GOLD [XAUT]	$ 430M	+6.2%
12	EURO TETHER [EURT]	$ 243M	+3.8%
13	TERRA CLASSIC USD [USTC]	$ 200M	-4.3%
14	LIQUITY USD [LUSD]	$ 185M	-0.5%
15	STASIS EURO [EURS]	$ 122M	-0.8%

CryptoRank.io @CryptoRank_io @CryptoRankNews

Data source: CryptoRank.io Nov 28, 2022 CRYPTORANK

MINING REVENUE HAS SIGNIFICANTLY DROPPED OVER THE PAST YEAR

	COIN	1Y MINER REVENUE CHANGE	1Y PRICE CHANGE	MARKET CAP	RECENT DAILY MINING REVENUE	1Y MINER REVENUE CHART
1	BITCOIN [BTC]	+5.1%	-19.6%	$572B	$33.9M	
2	ZCASH [ZEC]	-18.7%	-36.0%	$1.16B	$306K	
3	ETHEREUM [ETH]	-43.6%	-31.4%	$217B	$23.2M	
4	MONERO [XMR]	-55.6%	-32.4%	$3.36B	$80.8K	
5	BITCOIN GOLD [BTG]	-59.9%	-69.2%	$381M	$19.6K	
6	LITECOIN [LTC]	-64.0%	-64.7%	$4.28B	$407K	
7	BITCOIN SV [BSV]	-65.3%	-67.5%	$1.08B	$53.2K	
8	ETHEREUM CLASSIC [ETC]	-68.6%	-65.7%	$2.87B	$362K	
9	BITCOIN CASH [BCH]	-69.0%	-72.3%	$3.36B	$168K	
10	DASH [DASH]	-71.4%	-67.4%	$619M	$78.2K	
11	DOGECOIN [DOGE]	-76.2%	-77.0%	$10.5B	$1.07M	
12	GRIN [GRIN]	-79.5%	-77.8%	$8.37M	$6.37K	
13	DIGIBYTE [DGB]	-83.2%	-81.6%	$186M	$31.1K	
14	VERTCOIN [VTC]	-84.9%	-71.6%	$11.2M	$1.22K	
15	VERGE [XVG]	-96.0%	-86.6%	$68.2M	$298	

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Bitcoin

Bitcoin: From Cypherpunk to Institutional Asset

Bitcoin as Money

- **Oldest and largest**: $1.3T+ market cap (2024); 21 million coin hard cap
- **Payments**: 100,000+ merchants accept Bitcoin; El Salvador made it legal tender (2021)
- **Scalability**: base layer 7 TPS; Lightning Network enables near-instant micropayments off-chain
- **CME Bitcoin Futures**: launched Dec 2017; open interest $10B+ in 2024

Bitcoin Cycles and Halvings

- **Halving**: block reward cut 50% every 4 years; April 2024 reduced reward to 3.125 BTC
- **Price pattern**: each halving historically preceded a major bull run (2013, 2017, 2021, 2024)

Institutional Adoption

- **MicroStrategy**: holds 214,000+ BTC ($14B+); CEO Michael Saylor pioneered corporate Bitcoin treasury strategy
- **BlackRock IBIT ETF**: approved Jan 2024; $10B AUM in 49 days – fastest ETF in history
- **Coinbase**: NASDAQ-listed; official custodian for 8 of 11 Bitcoin spot ETFs
- **Tesla**: held $1.5B in Bitcoin (2021); partially sold; still holds position

Key Lesson

Bitcoin spot ETF approval (Jan 2024) was a watershed: it opened Bitcoin to pension funds, RIAs, and retail brokerage accounts – fundamentally changing the demand structure.

Lightning Network: Scaling Bitcoin Payments

How Lightning Works

- **Payment channels**: two parties lock funds in a multi-sig address; no on-chain fees per transaction
- **Off-chain transactions**: unlimited transfers at near-zero cost and millisecond speed
- **Settlement**: when finished, one transaction settles net balance on-chain
- **Routing**: payments route through multiple channels without a direct channel required

Real-World Adoption

- **El Salvador**: Lightning built into Chivo wallet for national payments
- **Strike app**: 13M+ users; integrates with Twitter/X tips via Lightning

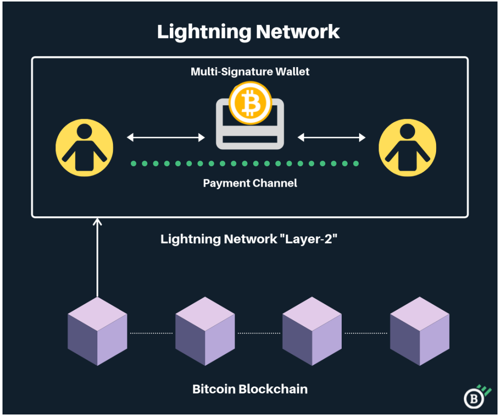
SegWit: The Foundation

- **Segregated Witness (SegWit)**: separates signature data from transaction data; increases effective block size from 1MB to ~1.8MB
- **Bitcoin Cash (BCH)**: hard fork Aug 2017 by miners preferring 8MB blocks instead of SegWit
- **Bitcoin dominance**: SegWit adoption now 85%+ of Bitcoin transactions

Key Lesson

Lightning solves Bitcoin's throughput problem: Visa handles 24,000 TPS; Bitcoin base layer 7 TPS; Lightning theoretically millions of TPS. The tradeoff is liquidity management and routing complexity.

Lightning Network: Architecture



Bitcoin Mining: Economics and Energy

Mining Economics

- **Hashrate**: Bitcoin network at 620 EH/s (2024 all-time high)
- **ASIC manufacturers**: Bitmain (Antminer S21: 200 TH/s) and MicroBT dominate hardware
- **Mining revenue**: miners earned \$10B+ in 2023 (block reward plus transaction fees)
- **Profitability**: depends on BTC price, electricity cost, hardware efficiency

Geographic Shifts

- **China ban (2021)**: was 65% of hashrate; overnight shift to US, Kazakhstan, Russia
- **US dominance**: Texas, Kentucky, Georgia now lead; 38% of global hashrate

Energy Debate

- **Annual energy**: Bitcoin uses ~150 TWh/year – comparable to Poland's electricity consumption
- **Renewable mix**: Cambridge Centre estimates 50%+ renewable energy in Bitcoin mining
- **Stranded energy**: miners use excess hydropower (Sichuan), wind (Texas), natural gas flaring
- **Criticism**: ESG concerns from institutional investors; Tesla halted BTC payments (2021)

Key Lesson

Mining is a competitive industry: when BTC price rises, more hashrate joins; difficulty adjusts every 2,016 blocks (≈2 weeks) to maintain 10-minute blocks – an elegant self-regulating mechanism.

Ethereum

Ethereum: The Programmable Blockchain

Founded: Vitalik Buterin (2014); ICO raised \$18.4M in Bitcoin – history's most successful fundraise

Smart contracts: Turing-complete programs that execute automatically on-chain; no middlemen required

The Merge (Sep 2022): switched from Proof-of-Work to Proof-of-Stake; reduced energy consumption by 99.95%

ETH staking yield: validators earn 3–4% APY by locking 32 ETH; \$60B+ total staked as of 2024

EIP-1559 (2021): base fee burned with each transaction; made ETH deflationary – 1.2M+ ETH burned in 2023

Ethereum Ecosystem: Layer 2 and DeFi

Layer 2 Scaling Solutions

- **Arbitrum**: largest L2 by TVL ($18B+); optimistic rollups; 40,000 TPS theoretical
- **Optimism / Base**: Coinbase launched Base on Optimism stack (2023); 2M+ daily active addresses
- **zkSync and Starknet**: zero-knowledge rollups; cryptographically prove correctness
- **Combined L2 TVL**: $40B+ locked across all Ethereum L2s (2024)

The DAO Attack (2016)

- **Exploit**: hacker drained $60M from DAO smart contract via re-entrancy bug
- **Hard fork**: community voted to reverse; created Ethereum (ETH) and Ethereum Classic (ETC)

Proof-of-Work vs. Proof-of-Stake

- **PoW**: miners compete solving hash puzzles; energy-intensive; hardware-farm advantage
- **PoS**: validators stake ETH as collateral; randomly selected; slashing penalizes misbehavior
- **Nothing-at-stake**: Casper protocol solves by slashing stake for signing multiple forks
- **Energy**: Ethereum post-Merge uses 0.01 TWh/year vs. 78 TWh pre-Merge

Key Lesson

Ethereum is the settlement layer for DeFi, NFTs, and stablecoins. Layer 2s inherit its security while delivering speed and low fees – the model for scalable blockchain infrastructure.

Proof of Work vs. Proof of Stake: Comparison

Proof of Stake	Proof of Work
Block creators are called validators	Block creators are called miners
Participants must own coins or tokens to become validators	Participants must buy equipment and energy to become a miner
Energy efficient	Not energy efficient
Security through community control	Robust security due to expensive upfront requirement
Validators receive transaction fees as rewards	Miners receive block rewards and fees

Altcoins and Stablecoins

Ripple/XRP: Cross-Border Payments

RippleNet: payment network targeting global bank-to-bank transfers; competes with SWIFT for cross-border settlements

XRP: native token; all 100 billion pre-mined at launch (2012); not created through mining

Speed: XRP settlement in 3–5 seconds at fraction of a cent; vs. SWIFT's 1–5 business days

SEC lawsuit: filed Dec 2020 alleging XRP is unregistered security; partial victory for Ripple in 2023 – XRP not a security in programmatic sales; case ongoing for institutional sales

Bank partners: Santander, SBI Holdings, PNC Bank among 300+ financial institutions

Ripple: XRP Ledger and Consensus

XRP Ledger protocol: no mining; validators use federated Byzantine Agreement (FBA)

Transaction flow: bank deposits funds to RippleNet partner; converted to XRP; sent globally; converted to local currency in seconds

Validators: trusted nodes maintain Unique Node List (UNL); transactions public but amounts confidential

Competition with SWIFT: SWIFT handles $5T/day in messages; Ripple targets same market with real-time finality and lower cost

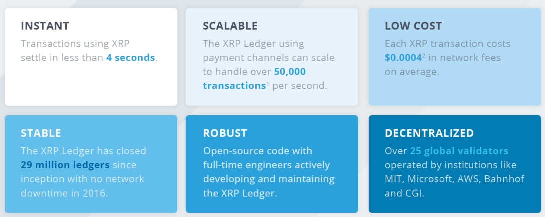
Ripple: Validator Incentives

Validators decide the evolution of the XRP Ledger; participants have inherent incentive to ensure reliability and stability

Reputation-based: validators earn trust and goodwill of the community by contributing to consensus

No block reward: unlike Bitcoin miners, XRP validators are not paid in new coins; incentive is network utility and transaction throughput

Escrow: Ripple holds 55B XRP in escrow (released 1B/month); ensures predictable supply



Solana, Litecoin and Altcoin Landscape

Solana: High-Speed Blockchain

Speed: 65,000 TPS theoretical; 3,000–4,000 TPS in production; sub-$0.001 fees

Proof of History: cryptographic clock timestamps transactions before consensus

Visa partnership: USDC settlement expanded to Solana (2023)

Outages: multiple outages (2022); performance vs. decentralization tradeoff

Litecoin (LTC)

Founded: Charlie Lee (ex-Google), Oct 2011; 84M coin cap (4x Bitcoin)

Speed: 2.5-minute blocks vs. Bitcoin's 10; Scrypt resists ASIC dominance

BNB and Binance Ecosystem

BNB: native token of Binance; 25% trading fee discount; BNB Smart Chain gas

Quarterly burn: Binance burns BNB using 20% of profits; 50M BNB burned (2024)

Interoperability Coins

Polkadot: parachains connect blockchains via relay chain; enables cross-chain transfers

Cardano: peer-reviewed academic approach; Proof-of-Stake (Ouroboros); 5M+ wallets

Key Lesson

The altcoin market is winner-take-most: Ethereum dominates smart contracts; Solana challenges on speed; Ripple targets payments. Most altcoins fail – Bitcoin and Ethereum capture 60%+ of total crypto value.

Litecoin: Key Technical Features

Founded: Charlie Lee (former Google engineer); launched Oct 7, 2011; 84M LTC total supply

Block time: 2.5 minutes vs. Bitcoin's 10 minutes – 4x faster confirmation

Algorithm: Scrypt (memory-hard function by Colin Percival); designed to resist ASIC dominance

SegWit: Litecoin adopted SegWit in May 2017 – two months before Bitcoin; proved the technology viable

Lightning Network: Litecoin also supports Lightning; first cross-chain atomic swap with Bitcoin (2017)

Dash: Two-Tier Governance and Privacy

Origin: "Digital Cash"; launched Jan 2014; forked from Litecoin; previously named Darkcoin

Masternodes: require 1,000 DASH collateral; perform advanced functions and vote on proposals; reward split: 45% miners, 45% masternodes, 10% treasury

PrivateSend: CoinJoin-based mixing of transaction addresses; only available in fixed denominations for untraceability

InstantSend: inputs locked and verified by masternode consensus in <2 seconds; competes with credit card speed

Monero: Privacy by Design

Launched: April 18, 2014; privacy is default, not optional

Ring signatures: mixes spender's address with a group of others – sender untraceable

Stealth addresses: recipient generates one-time address per transaction – receiver untraceable

RingCT: conceals transaction amounts; all three privacy properties combined

Regulatory pressure: Coinbase, Kraken delisted XMR; classified as high-risk by FATF; used by WannaCry ransomware operators to launder proceeds

Stablecoins: The $140B+ Market

Fiat-Backed Stablecoins

- **USDT (Tether)**: largest stablecoin; $90B+ supply; backed by USD reserves and T-bills
- **USDC (Circle)**: $30B+ supply; monthly Deloitte attestations; preferred by institutions
- **Total market**: $140B+ (2024); used for trading, remittances, and DeFi collateral
- **Regulatory push**: US Stablecoin Act 2024 would require full reserves and federal licensing

Algorithmic Failure: TerraUSD

- **Mechanism**: UST peg maintained via algorithmic arbitrage with LUNA token
- **Collapse (May 2022)**: depeg spiral destroyed $40B in 72 hours; LUNA fell from $120 to $0.0001

How Algorithmic Pegs Break

- **Bank run dynamics**: when confidence breaks, everyone redeems at once; death spiral inevitable
- **Anchor Protocol**: offered 20% APY on UST deposits — attracted $14B before collapse
- **Contagion**: Three Arrows Capital, Celsius, Voyager all failed within weeks
- **Do Kwon**: TerraUSD founder; arrested in Montenegro (2023); extradited to US

Key Lesson

Stablecoins are not all equal: fiat-backed (USDC) carries issuer risk; algorithmic (UST) carries death-spiral risk. The $40B Terra collapse proved algorithmic stablecoins cannot survive a bank run without genuine collateral.

Zcash: Selective Privacy

Launched: Oct 2016; fixed supply of 21 million ZEC (same as Bitcoin)

Zero-knowledge proofs (zk-SNARKs): mathematically prove a transaction is valid without revealing sender, recipient, or amount

Shielded vs. transparent: users choose private (z-addresses) or public (t-addresses) transactions on same network

Selective disclosure: share viewing key for tax audits or compliance – meets AML requirements while preserving privacy option

Regulation: delisted from Coinbase UK and several European exchanges under Travel Rule pressure

ICOs and Token Economics

ICO Era: Fundraising Revolution and Collapse

The ICO Boom (2017–2018)

Definition: ICO raises funds by selling new cryptocurrency tokens, typically for ETH or BTC

Scale: $7B raised in 2017; $21B in 2018; dominated blockchain startup financing

White paper: companies publish project vision, tokenomics, team – replaces prospectus

Access: anyone globally could invest; no accredited investor requirements

ICO vs. IPO: tokens convey utility rights; shares convey ownership; different legal frameworks

ICO Failure Rate

80%+ failed: Satis Group (2018) found 80% of 2017 ICOs were scams or worthless within 1 year

Exit scams: teams raised funds, disappeared with ETH; no product ever built

Fraud examples: Centra Tech ($25M, celebrity-backed) – founders arrested; BitConnect ($2.5B Ponzi) – collapsed 2018

Regulatory crackdown: SEC charged 75+ projects; China and South Korea banned ICOs (Sept 2017)

Key Lesson

ICOs democratized fundraising but failed on investor protection. The ease of raising capital with no accountability created the largest fraud wave in tech history.

ICO vs. IPO and Other Fundraising Mechanisms

	IPO	ICO
You buy	Shares	Tokens
Launched by	Well-established organisations	Startups (mainly)
Regulations	Well regulated by various regulators	Lacking any regulations
Working product	Based on past record or available	Very rare

IPOs: list company shares on stock exchange; SEC registration; audited financials; investor protections

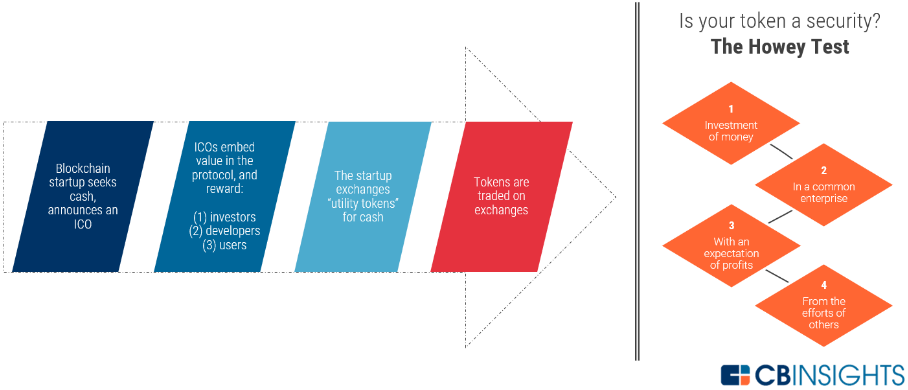
ICOs: sell cryptocurrency tokens; minimal disclosure; global access; high fraud risk; 80%+ failure rate

IEOs: token sales conducted on exchanges (Binance Launchpad); exchange vets projects, provides legitimacy and instant liquidity

STOs: Security Token Offerings; tokens registered as securities; full investor protections; smaller but compliant; Overstock tZERO raised $134M

DeFi launches: UNI (Uniswap) distributed 400 UNI free to 250,000 past users – worth $1,400 per wallet at launch

What's an ICO?

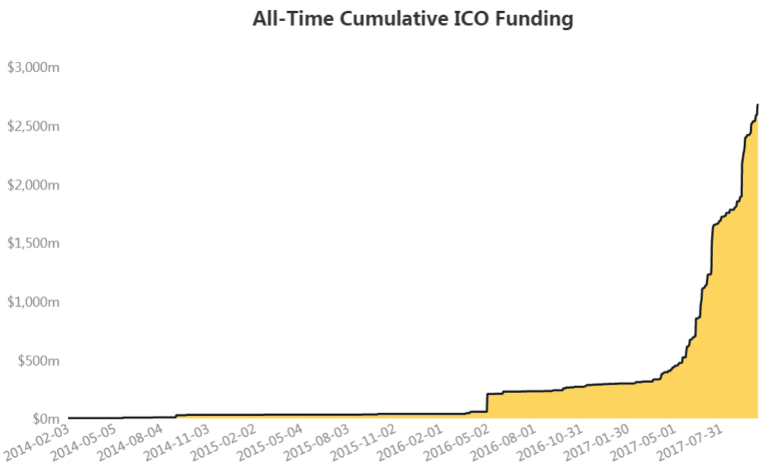

Blockchain startup seeks cash, announces an ICO
ICOs embed value in the protocol, and reward:
(1) investors
(2) developers
(3) users
The startup exchanges "utility tokens" for cash
Tokens are traded on exchanges
Is your token a security?
The Howey Test
1
Investment of money
2
In a common enterprise
3
With an expectation of profits
4
From the efforts of others
CBINSIGHTS

Returns of ICOs as of 2017

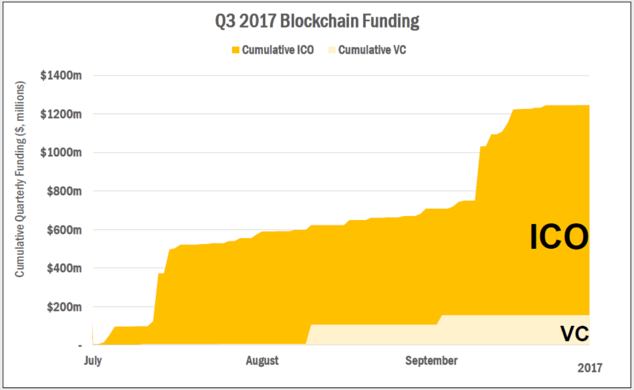
Asset:	Ethereum	Augur	Golem	Iconomi	Melonport	First Blood	Digix	SingularDTV
Website:								
Token:	ETH	REP	GNT	ICN	MLN	1ST	DGD	SNGLS
Market Cap:	$28.16bn	$295m	$427m	$360m	$44m	$156m	$166m	$106m
Price:	$294.92	$28.10	$.48	$3.85	$71.66	$1.66	$80.78	$.17
ICO Price:	$.31	$.60	$.01	$.13	$5.80	$.06	$3.24	$.015
ICO Date:	9/2/2014	8/1/2015	11/11/16	8/25/2016	2/15/17	9/25/2016	3/1/2016	10/5/2016
Return:	95,035%	4,583%	4,700%	2,862%	1,136%	2,667%	2,393%	1,033%

Data Source: Coinmarketcap, ICOstats
Notes: Market cap and price as of Q2 2017 (6/30/17), ICO stats calculated per $ prices during ICO dates, not all ETH dapps shown

coindesk


All-Time Cumulative ICO Funding
$3,000m
$2,500m
$2,000m
$1,500m
$1,000m
$500m
$0m
2014-02-03
2014-05-05
2014-08-04
2014-11-03
2015-02-02
2015-05-04
2015-08-03
2015-11-02
2016-02-01
2016-05-02
2016-08-01
2016-10-31
2017-01-30
2017-05-01
2017-07-31

ICO Funding Far Exceeded VC Funding for Blockchain Companies in 2017



ICO
$1,246m

VC
$156m

Top ICO Deals:
Filecoin - $262m
Tezos - $232m
TenX - $83.1m
MobileGo - $53.1m

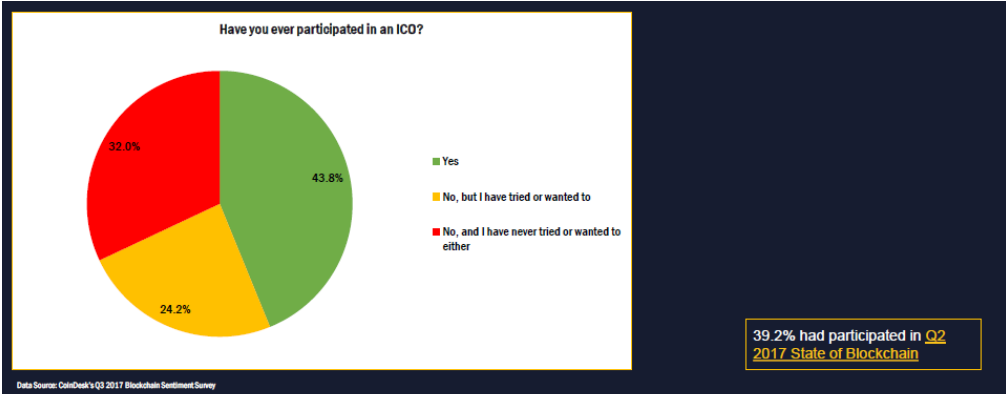
Top VC Deals:
Coinbase - $100m
Bitmain - $50m
Conjoule - $4.5m
Bitcoin.com.au - $0.8m

Q1	
ICO $36m	VC $107m

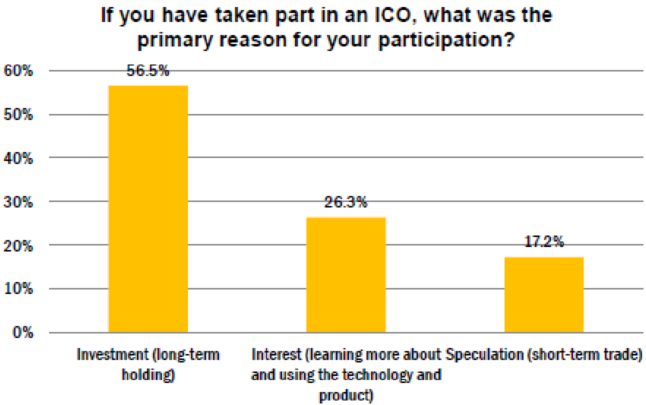
Q2	
ICO $797m	VC $235m

Data Sources: CoinDesk venture capital database, CoinDesk ICO Tracker

Participation in the ICO

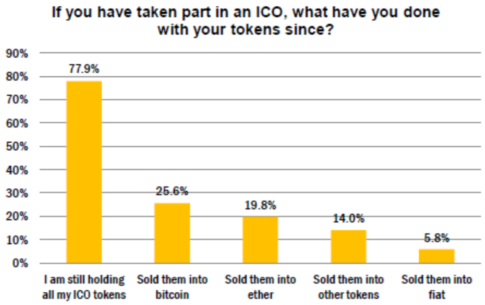


Reasons for ICOs



Source: CoinDesk

What to Do with the Tokens?



Source: CoinDesk


134 blockchain startups with ICOs
Closed initial coin offerings greater than or equal to $500K. 2014 – 2017 (09/08/2017)
ASSET MANAGEMENT
EXCHANGES & WALLETS
BROWSERS & SOCIAL
CROWDFUNDING & LENDING
MEDIA & ADVERTISING
TRADING
PREDICTION MARKETS
GAMBLING & GAMING
HEALTHCARE & INSURANCE
DAO & TOKEN LAUNCH
IDENTITY & INTERNET OF THINGS
COMPUTING & STORAGE
PAYMENTS & BANKING
INFRASTRUCTURE & DEVELOPMENT
OTHER
FINANCIAL SERVICES
Sources: CB Insights, TokenData, CoinSchedule. Map is illustrative; not exhaustive.
We make no claims to the value of any of these projects.
CBINSIGHTS

Token Economics: Design Matters

Utility vs. Security Tokens

- **Utility token**: grants access to a service; not a security if no profit expectation (e.g., Filecoin for storage)
- **Security token**: represents ownership or profit rights; must register with SEC; Howey Test applies
- **Howey Test**: investment of money + common enterprise + profit expectation + from efforts of others = security
- **Most ICOs were securities**: tokens sold with profit expectation qualify regardless of "utility" label

Vesting and Distribution

- **Team allocation**: typically 15–20%; locked 1–4 years to align incentives
- **Investor allocation**: 10–30%; early investors receive largest discount

Tokenomics Mechanics

- **Inflation**: new tokens minted to reward validators; excessive inflation dilutes holders
- **Burn**: tokens permanently destroyed; ETH burns 1M+ per year; BNB burns 20% of Binance profits
- **Staking rewards**: lock tokens to earn yield; reduces supply; creates price support
- **Curve Wars**: DeFi protocols bribe CRV voters to direct liquidity; governance token became geopolitical tool

Key Lesson

Tokenomics is a new field of mechanism design. Well-designed supply schedules, burn mechanisms, and staking incentives create sustainable ecosystems. Poorly designed tokenomics (high inflation, no utility) guarantee failure.

Ethereum: History's Most Successful ICO

Date: July 22, 2014; raised \$18.4 million in Bitcoin over 42 days

Price: 2,000 ETH per BTC in first 14 days; 1,337 ETH per BTC thereafter

Return: ETH ICO investors paid \$0.31/ETH; all-time high \$4,891 (Nov 2021) – 15,000x return

Use of funds: development, legal, marketing; Vitalik Buterin and team delivered working mainnet in July 2015

Contrast: EOS raised \$4.1B over 1 year (2017–2018); Telegram GRAM raised \$1.7B but SEC blocked launch, returned funds

Source: Ethereum.org

- Ether is a product, **NOT a security or investment offering.** Ether is simply a token useful for paying transaction fees or building or purchasing decentralized application services on the Ethereum platform; it does not give you voting rights over anything, and we make no guarantees of its future value.
- The price of ether is initially set to a discounted price of **2000 ETH per BTC**, and will stay this way for **14 days** before linearly declining to a final rate of **1337 ETH per BTC**. The sale will last **42 days**, concluding at 23:59 Zug time September 2.

BAT: The Fastest ICO – $36M in 24 Seconds

Brave Browser: open-source, privacy-focused browser blocks ads and trackers by default; replaces ad ecosystem with crypto micropayments

Basic Attention Token (BAT): May 31, 2017; raised $36M (156,250 ETH) in just 24 seconds on Ethereum

Mechanism: users earn BAT for viewing privacy-respecting ads; publishers receive BAT based on verified attention

Current scale: Brave has 70M+ monthly active users (2024); 1.6M verified publishers; BAT ecosystem processes $50M+ in monthly creator payments

The DAO: Most Tragic ICO – $60M Hacked

What Was the DAO?

Decentralized Autonomous Organization: first on-chain venture fund; raised $152M in ETH (May 2016) – largest crowdfund in history

Governance: token holders voted on investments via smart contract; no human managers needed

Vision: autonomous, trustless capital allocation without intermediaries

The Hack

Re-entrancy bug: attacker drained $60M by calling withdraw recursively before balance updated

Community response: hard fork reversed all DAO transactions — violated blockchain immutability principle

Ethereum Classic: supporters of the original chain split off; ETC still trades today

SEC Consequences

SEC report (July 2017): DAO tokens were securities under Howey Test; no enforcement but clear warning

Precedent: on-chain governance tokens can be securities regardless of technical structure

Delisted: DAO tokens removed from major exchanges in late 2016; effectively worthless

Key Lesson

Code is law – until $60M is stolen. The DAO hack demonstrated that smart contract bugs can be catastrophic and irreversible. The hard fork decision split the Ethereum community over core philosophical principles.

ICO Case Studies: Unusual Examples

MilkCoin (2017): Agricultural ICO

- **Structure**: tokenized milk complex in Voronezh, Russia; 2,400 cows; 3,500 hectares
- **Token (MLCN)**: annual dividends; company can repurchase at 20% premium
- **Takeaway**: predated the 2024 Real World Asset (RWA) tokenization trend by 7 years

BetBox (2017): Sports Betting Hedge Fund

- **Model**: algorithmic sports betting; 3% management fee plus 30% performance fee
- **Token**: BETX holders receive 20% of net transaction revenue
- **Lesson**: unregulated gambling plus unregistered securities – regulatory nightmare

BBOD: Decentralized Derivatives

- **Concept**: world's first decentralized crypto options and futures platform
- **Token burn**: all BBD fees permanently burned – deflationary model
- **Disclaimer**: "We do not recommend buying BBD tokens for speculation" – rare ICO-era transparency

XRED: Real Estate ICO

- **Target**: UK and Central London real estate; tokenized property investment
- **Value prop**: removes intermediary chains; ICO-wallet for direct control

Key Lesson

The 2017 ICO market revealed both genuine innovation (asset tokenization, decentralized derivatives) and blatant fraud. Today's tokenized RWA market ($5B+ in 2024) builds on legitimate ICO-era ideas with proper

Is an ICO a Security? The Howey Test

The Howey Test (SEC v. W.J. Howey, 1946)

- **Investment of money**: buyers spend fiat or crypto to purchase tokens — yes
- **In a common enterprise**: token value tied to shared project success — usually yes
- **Expectation of profits**: marketed as investment with return potential — yes for most ICOs
- **From efforts of others**: value created by the development team, not token holders — yes
- **Result**: most ICOs are securities; most issuers failed to register

SEC Enforcement Actions

- **DAO report (2017)**: first official guidance; no action but clear signal
- **Telegram GRAM**: SEC halted $1.7B ICO (2019); $18.5M settlement; funds returned
- **75+ charged**: Centra Tech, Longfin Corp, Kik's Kin token ($100M ICO, $5M penalty)
- **Ripple (2020)**: SEC sued over XRP; partial Ripple win 2023
- **Gary Gensler (2021–2024)**: 100+ crypto enforcement actions

Key Lesson

The Howey Test is 80 years old but applies perfectly to most tokens. Calling something a "utility token" does not make it one – the SEC looks at economic substance, not labels.

SEC Investigation of DAO

SEC Issues Investigative Report Concluding DAO Tokens, a Digital Asset, Were Securities

U.S. Securities Laws May Apply to Offers, Sales, and Trading of Interests in Virtual Organizations

FOR IMMEDIATE RELEASE
2017-131

Washington D.C., July 25, 2017 — The Securities and Exchange Commission issued an investigative report today cautioning market participants that offers and sales of digital assets by "virtual" organizations are subject to the requirements of the federal securities laws. Such offers and sales, conducted by organizations using distributed ledger or blockchain technology, have been referred to, among other things, as "Initial Coin Offerings" or "Token Sales." Whether a particular investment transaction involves the offer or sale of a security – regardless of the terminology or technology used – will depend on the facts and circumstances, including the economic realities of the transaction.

SEC Investigation of DAO: Key Documents

projects in exchange for a return on investment.[35] The ETH was pooled and available to The DAO to fund projects. The projects (or "contracts") would be proposed by Contractors. If the proposed contracts were whitelisted by Curators, DAO Token holders could vote on whether The DAO should fund the proposed contracts. Depending on the terms of each particular contract, DAO Token holders stood to share in potential profits from the contracts. Thus, a reasonable investor would have been motivated, at least in part, by the prospect of profits on their investment of ETH in The DAO.

Investors who purchased DAO Tokens were investing in a common enterprise and reasonably expected to earn profits through that enterprise when they sent ETH to The DAO's Ethereum Blockchain address in exchange for DAO Tokens. "[P]rofits" include "dividends, other periodic payments, or the increased value of the investment." *Edwards*, 540 U.S. at 394. As described above, the various promotional materials disseminated by Slock.it and its co-founders informed investors that The DAO was a for-profit entity whose objective was to fund

SEC Regulation of ICO

SEC Exposes Two Initial Coin Offerings Purportedly Backed by Real Estate and Diamonds

FOR IMMEDIATE RELEASE
2017-185

Washington D.C., Sept. 29, 2017 — The Securities and Exchange Commission today charged a businessman and two companies with defrauding investors in a pair of so-called initial coin offerings (ICOs) purportedly backed by investments in real estate and diamonds.

The SEC alleges that Maksim Zaslavskiy and his companies have been selling unregistered securities, and the digital tokens or coins being peddled don't really exist. According to the SEC's complaint, investors in REcoin Group Foundation and DRC World (also known as Diamond Reserve Club) have been told they can expect sizeable returns from the companies' operations when neither has any real operations.

Another SEC Enforcement Case

SEC Halts Alleged Initial Coin Offering Scam

FOR IMMEDIATE RELEASE
2018-8

Washington D.C., Jan. 30, 2018 — The Securities and Exchange Commission obtained a court order halting an allegedly fraudulent initial coin offering (ICO) that targeted retail investors to fund what it claimed to be the world's first "decentralized bank."

According to the SEC's complaint, filed in federal district court in Dallas on Jan. 25 and unsealed late yesterday, Dallas-based AriseBank used social media, a celebrity endorsement, and other wide dissemination tactics to raise what it claims to be $600 million of its $1 billion goal in just two months.

The SEC alleges that AriseBank falsely stated that it purchased an FDIC-insured bank which enabled it to offer customers FDIC-insured accounts and that it also offered customers the ability to obtain an AriseBank-branded VISA card to spend any of the 700-plus cryptocurrencies. AriseBank also allegedly omitted to disclose the criminal background of key executives.

International Regulation of ICOs

China has banned ICOs

Posted Sep 4, 2017 by Jon Russell (@jonrussell)

A notice from a committee led by China's central bank [link in Chinese] today announced an immediate ban on ICO funding, which has "seriously disrupted the economic and financial order."

Financial news site Caixin reported [link in Chinese] reported that the committee has prepared a list of 60 exchanges which will be subject to inspection and a report. In the meantime, there will be an ICO freeze in China.

ICOs involve raising funding by creating and selling new crypto tokens — commonly based on Ethereum — to investors. That's led to comparisons with securities, with much speculation over whether financial regulators will look to regulate the space.

The Chinese committee voiced concern that some ICOs are financial scams and pyramid schemes. That echoes a recent warning from Singapore's MAS.

Source: TechCrunch

International Regulation of ICOs: Country Approaches

"Raising funds through ICOs seem to be on the rise globally, and our assessment is that ICOs are increasing in South Korea as well," the regulator said in a statement after a meeting with the finance ministry, the Bank of Korea and the National Tax Service.

"Stern penalties" will be issued on financial institutions and any parties involved in issuing of ICOs, the statement added, without elaborating further on the details of those penalties.

#TECHNOLOGY NEWS SEPTEMBER 28, 2017 / 10:46 PM / 4 MONTHS AGO

South Korea bans raising money through initial coin offerings

SEOUL (Reuters) - South Korea's financial regulator on Friday said it will ban raising money through all forms of virtual currencies, a move that follows similar restrictions in China on initial coin offerings.

The Financial Services Commission said all kinds of initial coin offerings (ICO) will be banned as trading of virtual currencies needs to be tightly controlled and monitored.

ICO Market: Rise, Collapse and Evolution

The Boom and Bust

- **2017**: $7B raised; ICOs surpassed VC funding for blockchain; dominated by retail speculation
- **2018**: $21B raised; peak activity; most tokens lost 90%+ alongside Bitcoin's 84% crash
- **2019**: market collapsed; only $371M raised; regulatory crackdowns eliminated most issuers
- **80%+ scam rate**: fake teams, no products, coordinated pump-and-dump (Satis Group, 2018)

What Replaced ICOs?

- **STOs**: tokens registered as securities; full investor protections; compliant; growing in 2024
- **IEOs**: launched on exchanges (Binance Launchpad); exchange vets projects for legitimacy
- **DeFi token launches**: governance tokens (UNI, AAVE, COMP) via airdrops — no ICO structure
- **NFT fundraising**: project pre-sales via NFT mint (2021–2022 boom); most also collapsed

Key Lesson

ICOs democratized fundraising but failed on investor protection. The evolution toward STOs and regulated token frameworks reflects the market learning that compliance is not optional for sustainable capital markets.

DeFi Governance and NFTs

DeFi Governance Tokens: DAOs in Practice

How DAO Governance Works

- **Governance tokens**: UNI (Uniswap), AAVE, COMP, CRV (Curve); holders vote on protocol parameters, fees, upgrades
- **Voting power**: proportional to token holdings; 1 token = 1 vote in most protocols
- **Quorum**: typically 4% of supply must vote; average 5–15% turnout
- **Timelock**: approved proposals execute after 48-hour delay
- **Delegation**: token holders delegate votes to active participants

The Curve Wars: Governance as Competition

- **CRV token**: controls liquidity routing on Curve Finance ($5B TVL)
- **Convex Finance**: locked $10B in CRV to accumulate voting power; created meta-governance layer
- **Bribery**: protocols pay Votium bribes ($50M+) to CRV voters — explicit governance market
- **Concentration risk**: top 10 wallets control 60%+ of governance power — not truly decentralized

Key Lesson

DeFi governance tokens claimed to democratize finance but concentrated power in practice. The Curve Wars showed that governance rights have real economic value – and can be bought and sold like any commodity.

NFT Market: $25B Peak and 97% Collapse

NFT Boom (2021–2022)

- **Total NFT market**: peaked at $25B trading volume (2021); CryptoPunks and Bored Apes defined the era
- **BAYC**: 10,000 algorithmically generated apes; floor price peaked at $400K (April 2022); celebrities purchased
- **Beeple sale**: "Everydays: The First 5000 Days" sold at Christie's for $69.3M (March 2021)
- **OpenSea**: largest NFT marketplace; $3.4B monthly volume (Jan 2022)

NFT Collapse (2022–2023)

- **97% collapse**: NFT volume fell from $25B to $800M; most collections became worthless
- **BAYC floor price**: peaked $400K; fell to $30K by 2023 (92% decline)
- **Blur vs. OpenSea**: Blur captured 70%+ of volume via token incentives (2023)
- **Speculative vs. utility**: pure speculation collapsed; NFTs with real utility (gaming, ticketing) retain value

Key Lesson

NFTs introduced provable digital ownership – a genuine innovation. But 97% of NFT projects were purely speculative. The survivors have genuine utility: access rights, gaming assets, or brand membership.

Crypto Regulation

Global Crypto Regulation: Diverging Approaches

Comprehensive Frameworks

- **EU MiCA**: passed June 2023; effective Dec 2024; covers stablecoins, exchanges, token issuers; requires reserve backing and licensing
- **UAE**: crypto-friendly; VARA (Dubai) licenses exchanges and funds; Binance, Bybit headquartered there
- **Singapore MAS**: licensing regime since 2020; Crypto.com, Coinbase licensed
- **Japan FSA**: first to regulate crypto exchanges (2017 after Mt. Gox); strict but clear

Restrictive and Enforcement-Heavy

- **US SEC vs. CFTC**: jurisdictional battle; Bitcoin = commodity (CFTC); most tokens = securities (SEC); no comprehensive law (2024)
- **Gary Gensler**: 100+ SEC enforcement actions (2021–2024); sued Coinbase, Binance, Kraken, Ripple
- **China**: complete ban on crypto trading and mining (2021); e-CNY as state alternative
- **India**: 30% capital gains tax plus 1% TDS; effectively prohibitive

Key Lesson

Regulatory clarity drives institutional adoption. Countries with clear frameworks (EU, UAE, Singapore) attract crypto businesses; enforcement-heavy uncertainty (US) pushes activity offshore.

FTX Collapse: $8B Fraud and Its Consequences

What Happened

FTX: second-largest crypto exchange; valued at $32B (Jan 2022); SBF was crypto's public face

Fraud: FTX secretly transferred $8B in customer funds to sister firm Alameda Research

Collapse: CoinDesk revealed Alameda's FTT-heavy balance sheet (Nov 2022); Binance withdrew $500M; bank run; FTX bankrupt in days

Contagion: BlockFi, Genesis, Voyager also failed; $1T+ wiped from crypto market cap

Legal Aftermath

SBF arrest: Dec 2022 in Bahamas; extradited to US; convicted on 7 counts of fraud (Nov 2023)

Sentence: 25 years in federal prison (March 2024)

Customer recovery: FTX estate recovered $14.5B; full repayment planned at claim value

Regulatory impact: accelerated Congressional push for crypto legislation; proof-of-reserves became standard

Key Lesson

FTX showed that "decentralized" crypto was highly centralized in practice. Custodial exchanges with billions in unsegregated customer funds created the same risks as traditional financial fraud – just faster and larger.

Regulation of Cryptocurrencies: Historical Overview

Early Regulatory Positions

- **Australia**: Bitcoin classified as currency (GST applies); AML/CTF Act covers exchanges
- **Japan**: legal payment method since 2017 (Payment Services Act); 30%+ capital gains tax
- **United States**: IRS treats crypto as property; CFTC oversees Bitcoin futures; SEC claims most tokens are securities
- **Europe**: MiCA (2024) creates unified framework; previously fragmented by member state
- **China**: banned crypto-fiat exchanges (2017, 2021); promotes e-CNY instead

New York BitLicense

- **Issued by**: NY State DFS; first state-level crypto license (2015)
- **Required for**: receiving, transmitting, storing, or issuing virtual currency as a business in NY
- **Excluded**: software development; merchants accepting crypto; personal investment
- **Compliance burden**: $100K+ in legal costs; drove firms to incorporate elsewhere ("New York Exodus")

Key Lesson

Regulatory fragmentation imposes real costs: BitLicense's $100K+ compliance cost drove crypto firms to Wyoming, which created a crypto-friendly charter in 2019. Regulatory arbitrage shapes where blockchain innovation happens.

Central Bank Digital Currencies

Classical Central Bank Roles and CBDC Motivation

Traditional Central Bank Functions

- **Currency issuance**: issue and replace coins and paper currency; sole legal tender authority
- **Monetary policy**: buy/sell treasury bonds to adjust money supply and interest rates
- **Financial stability**: lender of last resort; regulate commercial banks; manage systemic risk
- **Payments oversight**: operate interbank settlement systems (Fedwire, TARGET2)

Why CBDCs?

- **Cash displacement**: Sweden uses cash for only 8% of retail transactions
- **Financial inclusion**: 1.4B unbanked adults; CBDC wallets accessible via mobile phone
- **Private stablecoin threat**: Facebook Libra (2019) alarmed central banks into accelerating CBDC research

Bank of England Research: CBDC Benefits

- **GDP boost**: CBDC issuance of 30% of GDP against government bonds could permanently raise GDP by 3%
- **Channels**: lower real interest rates; reduced distortionary taxes; lower monetary transaction costs
- **Settlement cost**: elimination of liquidity and credit risk between banks; 24/7 settlement around the clock
- **Structural risk**: retail deposits may shift to central bank; commercial banks lose funding base; bank disintermediation

Key Lesson

CBDCs are programmable central bank money. They can earn interest, expire, be geographically restricted, or automatically enforce tax. This programmability is both their greatest feature and biggest civil liberties concern.

RSCoin: Framework for Central Bank Cryptocurrency

Architecture: two-tier system; central bank at top; authorized commercial banks (“mintettes”) add transactions to low-level blocks

Mintette role: banks authorized by central bank verify and record transactions; exchange blocks with other mintettes

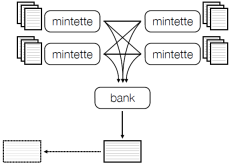
Central bank block: low-level blocks gathered and aggregated into high-level blocks; full auditability maintained

Advantage: inherits Bitcoin-style transparency without permissionless mining; central bank retains monetary control

Inspiration: influenced Bank of England’s CBDC research program; forerunner to e-CNY architecture

	CC	e-cash	Bitcoin	RSCoin
Double-spending	online	offline	online	online
Money generation	C	C	D	C
Ledger generation	C	n.a.	D	D*
Transparent	no	no	yes	yes
Pseudonymous	no	yes	yes	yes

C – centralized, D – decentralized, D - distributed*



Source: Centrally Banked Cryptocurrencies, UCL Research Paper

CBDC Global Status: Who Is Ahead?

Launched and Advanced

- **China e-CNY (DCEP)**: largest CBDC; 260M+ wallets; $250B+ in transactions; deployed in 26 cities; used at 2022 Winter Olympics; programmable – can expire, be geographically restricted
- **Bahamas Sand Dollar**: world's first CBDC (2020); small scale but fully operational
- **Nigeria eNaira (2021)**: low adoption (<0.5% population after 1 year); lessons on incentive design
- **India e-Rupee**: wholesale and retail pilots ongoing since 2022; RBI targeting full launch

Under Development

- **EU Digital Euro**: ECB evaluation phase; potential retail launch 2028; privacy vs. programmability debate ongoing; banks lobby against
- **US**: no CBDC; FedNow (2023) is a real-time payment rail, not a CBDC; strong political opposition to digital dollar
- **BIS mBridge**: multi-CBDC platform linking China, Hong Kong, Thailand, UAE – enables instant cross-border FX settlement; 134 countries (98% of world GDP) in CBDC exploration (2024 BIS Tracker)

Key Lesson

China's e-CNY is geopolitical: it challenges USD dominance in cross-border trade, enables sanctions evasion, and offers an alternative to SWIFT. The US refusal to develop a CBDC may be strategically costly.

Sweden: Pioneering the e-Krona

The case of Sweden

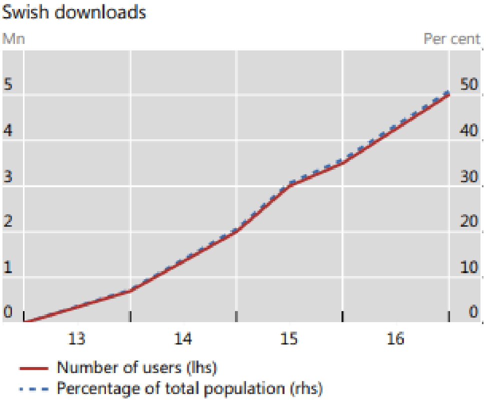
Sweden has one of the highest adoption rates of modern information and communication technologies in the world. It also has a highly efficient retail payment system. At the end of 2016, more than 5 million Swedes (over 50% of the population) had installed the Swish mobile phone app, which allows people to transfer commercial bank money with immediate effect (day or night) using their handheld device (Graph C, left-hand panel; see also Bech et al (2017)).

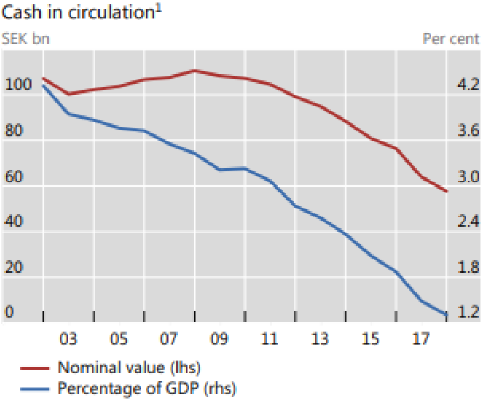
The demand for cash is dropping rapidly in Sweden (Graph C, right-hand panel). Already, many stores no longer accept cash and some bank branches no longer disburse or collect cash. These developments are a cause for concern for the Riksbank (Skingsley (2016)). Will the payment system continue to be safe and efficient without cash? Even if cash is not used every day, it is a backup option in crisis situations. Will those without access to bank services still be able to manage their payments?

The Riksbank currently has a so-called eKrona project under way to determine whether it should supply digital central bank money to the general public. The project is considering different technical solutions, but no decision has been taken as to whether to focus on a DCA or a retail CBCC structure. The project is expected to be finalised in late 2019 (Sveriges Riksbank (2017)).

Source: Bech and Garratt, Bank for International Settlements

Sweden: CBDC Design Options





Source: Bech and Garratt, Bank for International Settlements

Bank of England: CBDC Risks and Concerns

Currencies January 4, 2018

Bank halts crypto-currency plans over stability fears

FT ADVISER

Bank disintermediation: consumers would hold accounts directly at Bank of England; commercial banks lose deposit base and funding

Monetary policy effectiveness: demand for traditional money is elastic to interest rates; unclear if digital currency behavior would differ

Financial stability: transition risk – how to retrain staff, build infrastructure, manage bank balance sheet shrinkage

Current stance: Bank of England still studying; does not believe existing cryptocurrencies pose immediate systemic risk; "Britcoin" consultation ongoing

Algorithmic Stablecoins: Basecoin and TerraUSD Compared

Basecoin: The Prototype (2017)

Mechanism: peg to USD via algorithmic monetary policy; expand supply when price >$1; contract via bond auctions when price <$1

Bond-coin-share system: coins destroyed when buying bonds (contraction); bond holders paid first in expansion; shareholders receive remaining new coins

Backers: Andreessen Horowitz, Bain Capital Ventures, Pantera Capital – $133M raised

Outcome: shut down Dec 2018; SEC determined bonds and shares were unregistered securities; all funds returned

TerraUSD: The Disaster (2022)

Mechanism: UST peg maintained by arbitrage with LUNA token; mint 1 UST = burn $1 LUNA; redeem 1 UST = mint $1 LUNA

Anchor Protocol: 20% APY on UST deposits attracted $14B – unsustainable yield created illusion of safety

Collapse (May 2022): coordinated depeg attack triggered death spiral; $40B erased in 72 hours; LUNA from $120 to $0.0001

Contagion: Three Arrows Capital ($3B losses), Celsius (bankrupt), Voyager (bankrupt)

Key Lesson

Both Basecoin and TerraUSD shared the same fatal flaw: the peg depends on market confidence, not real collateral. Without hard assets backing every unit, algorithmic stablecoins are vulnerable to self-fulfilling bank runs.

China's Digital Yuan (e-CNY): CBDC at Scale

Architecture and Features

Centralized design: People's Bank of China issues e-CNY; distributed through authorized banks (ICBC, BOC, Alipay, WeChat Pay)

Controlled anonymity: small transactions pseudo-anonymous; large transactions linked to real identity; PBOC has full visibility

Programmability: can be programmed to expire, be geographically restricted, or restricted to specific merchants (e.g., COVID stimulus)

Scale: 260M+ wallets; $250B+ in transactions; 26 cities including Shenzhen, Shanghai, Beijing

Geopolitical Implications

Anti-money laundering: full transaction visibility enables real-time AML enforcement; eliminates cash-based evasion

RMB internationalization: USD holds 88%+ of international transactions; RMB only 4%; e-CNY targets cross-border trade settlement outside SWIFT

Financial inclusion: 200M+ unbanked Chinese citizens; e-CNY wallet requires only mobile phone, no bank account

Surveillance risk: programmable money enables unprecedented economic control; significant civil liberties concern

Key Lesson

China's e-CNY is the world's most advanced CBDC and a geopolitical tool. Its success demonstrates that digital sovereign currency works at scale – but its design prioritizes state control over user privacy.

References

Data Sources

CoinMarketCap: `coinmarketcap.com` – prices, market caps, historical data

CoinDesk: `coindesk.com` – ICO tracker and market analysis

Cambridge Centre for Alternative Finance: Bitcoin mining energy estimates (`ccaf.io`)

BIS CBDC Tracker: `cbdctracker.org` – 134 countries tracked

DeFiLlama: `defillama.com` – DeFi TVL and protocol data

Academic and Regulatory Sources

Satis Group (2018): Cryptoasset Market Coverage Initiation – 80% ICO scam finding

Broadridge and Garratt (2017): Centrally Banked Cryptocurrencies (UCL/BIS)

Bank of England: Macroeconomics of Central Bank Digital Currencies

SEC Enforcement: `sec.gov/spotlight/cybersecurity` – 75+ ICO actions

Investopedia: `investopedia.com` – concept definitions and examples